

The most important change was separating the runtime from the skills. In the prototype, `main.py` did everything. After the split, each request follows one fixed path (guardrail, route, permission check, execute, audit), and every skill only knows its own job. This made the system much easier to test, because we could test the runtime with fake skills and never start the model. Because each part now has one clear job, we can change or add a skill without touching the rest, the same governance and logging code is reused for every request, and failures show up as clear statuses.

The biggest technical challenge was running the tests on the bundled Windows Python. Our tests passed on macOS, but `Run Tests.cmd` failed with import errors. We found that the bundled Python uses a `python.pth` file, which puts it in isolated mode and ignores the working directory, so the test files could not import our packages. We fixed it by making each test file add the project root to `sys.path` itself. After that, all 38 tests passed.

I would make two improvements. First, replace keyword routing with embedding- or LLM-based routing, because requests without a listed keyword now fall through to unmatched. Second, add skill composition, so skills can chain together, for example knowledge lookup, then summary, then translation.